Investigating the Impact of REPA Loss on Training Speed and Generation Quality in Diffusion Models

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Motivation

REPA accelerates diffusion training by aligning a transformer's hidden states with clean features from a frozen encoder (DINOv2). We ask *when* this helps — and show a constant alignment weight eventually hurts.

The Problem: Gradient Conflict

We track the cosine between the diffusion and REPA gradients, $c = \cos(\nabla_w \mathcal{L}_{\text{diff}},\ \nabla_w \mathcal{L}_{\text{REPA}})$. It is *positive* at high noise but turns *negative* at low noise and late in training — there the auxiliary term pulls the update *against* the diffusion objective.

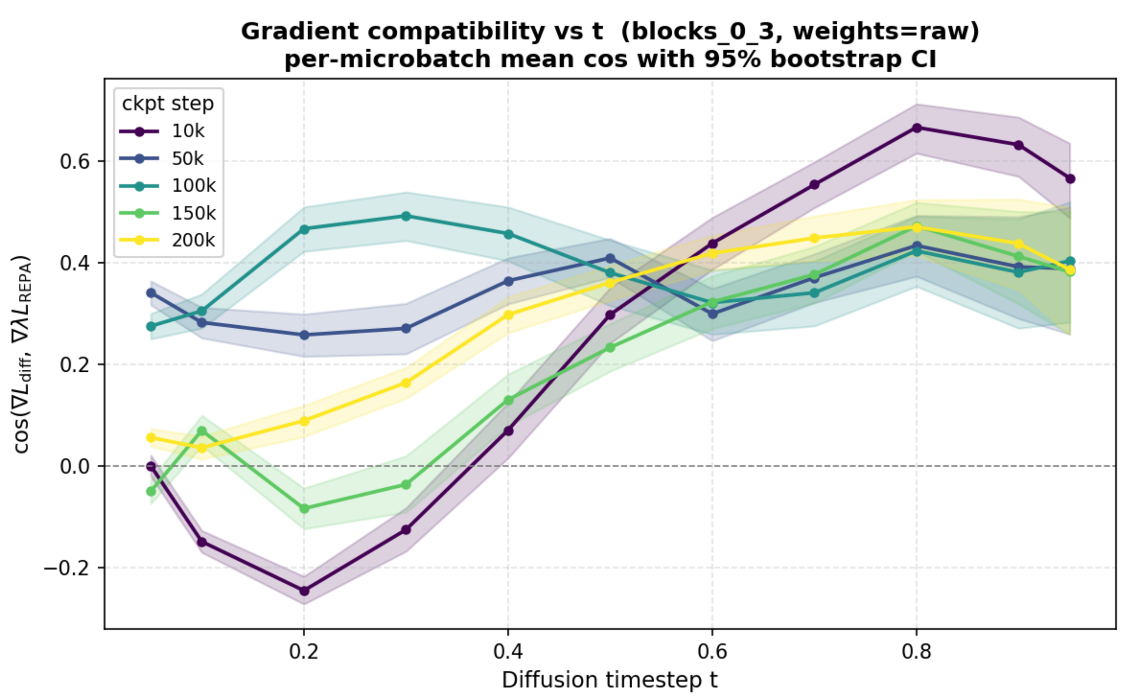

Figure 1. c vs. timestep t (high t = high noise). Note the negative region at low t.

REPA-Σ: Method & Guarantees

Discard *only* the part of the REPA gradient r anti-aligned with an EMA estimate $\bar{g}$ of the diffusion gradient g:

$$r^{\Sigma} = r - \frac{[\langle r, \bar{g}\rangle]_-}{\|\bar{g}\|_2^2}\,\bar{g}, \qquad [x]_- = \min(x, 0).$$

Minimal change: projection of r onto $\{r' : \langle r', \bar{g}\rangle \geq 0\}$.
Non-interference: $\langle g + \lambda r^{\Sigma}, \bar{g}\rangle \geq \langle g, \bar{g}\rangle$ for all $\lambda \geq 0$ — the adapted gradient never works against the diffusion descent direction. Parameter-free.

Toy Model: Whitened Linear Setting

Clean sample $x_0 \in \mathbb{R}^d$; whitened features u with $\mathbb{E}[uu^\top] = I$; linear denoiser $\hat{x}_0 = Bu$. The diffusion loss $L_{\text{diff}}(B) = \frac{1}{2}\mathbb{E}\|Bu - x_0\|^2$ has the unique optimum $A = \mathbb{E}[x_0 u^\top]$. A teacher projects onto an r-dim semantic subspace ($P^2 = P$, $Q = I - P$), giving $L_{\text{repa}}(B) = \frac{1}{2}\mathbb{E}\|Bu - Px_0\|^2$ — it keeps the semantic part Px_0 and *drops* the detail Qx_0.

Result. Standard REPA minimises $L_{\text{diff}} + \lambda L_{\text{repa}}$, with optimum $B^\star_\lambda = PA + \frac{1}{1+\lambda} QA$: the detail is shrunk, so it *cannot* reach A for $\lambda > 0$. REPA-Σ keeps A a stationary point for every λ — it provably reaches the optimal denoiser A while still using the alignment signal.

Toy Experiments

Experiments supporting the Toy Model results above — empirically confirming that REPA-Σ reaches the optimal denoiser A while vanilla REPA stays biased.

(1) Diagonal-matrix model with an independent-features teacher — closed-form convergence to A under REPA-Σ vs. biased REPA. (2) MLP denoiser on MNIST/CIFAR trained with a REPA-style teacher matching our setting.

Convergence across Three Datasets

This section compares the impact of **REPA** vs. **REPA**-Σ ($\pm\,\lambda$-anneal), against the diffusion baseline, on convergence speed (FID/KID) across three single-domain datasets spanning a spatial-difficulty axis. On **CelebA**, surgery helps mid-training and λ-annealing late; combining both gives the best final FID.

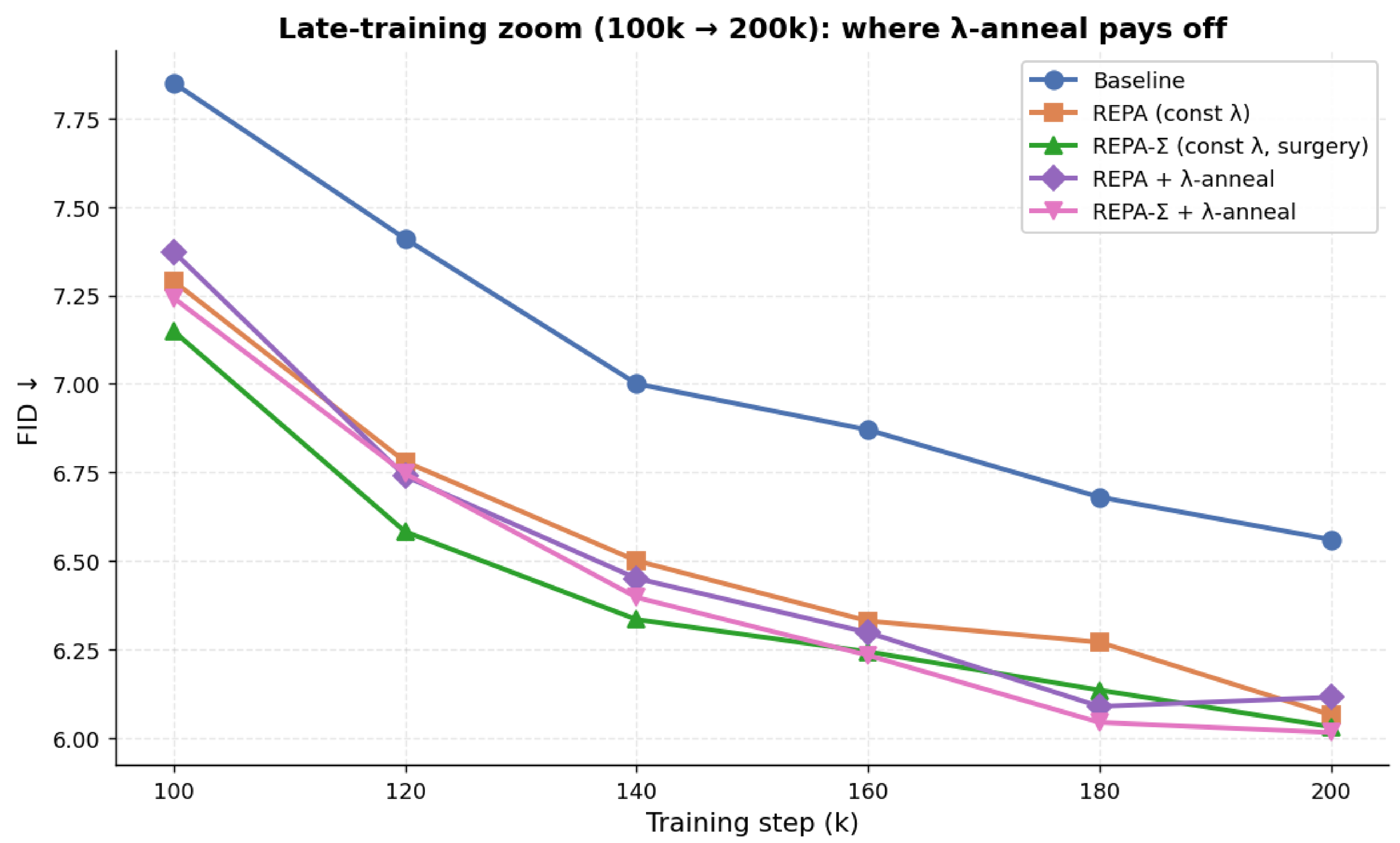

Figure 2. CelebA — late-training FID. REPA-Σ + λ-anneal wins.

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FID/KID convergence: baseline vs. REPA vs. REPA-Σ ($\pm\lambda$-anneal), plus the gradient-compatibility curve.

Comprehensive Cars (CompCars)

Same convergence comparison across all methods.

Feature Maps & Generation across Three Datasets

This section compares the impact of **REPA** vs. **REPA**-Σ on the learned feature maps and generations across the three datasets. PCA maps show REPA preserves semantic, face-centred structure through the mid-network and under heavy noise, where the baseline collapses; we extend the comparison to all three domains.

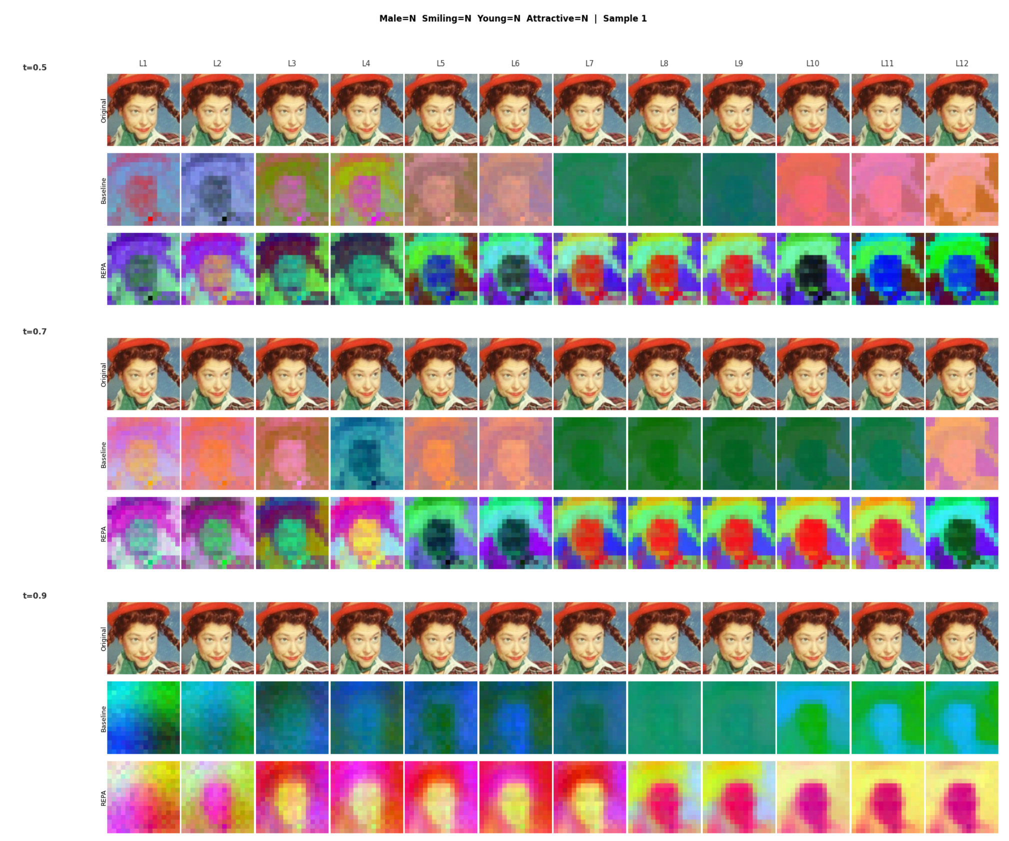

Figure 3. CelebA — per-block PCA maps: image / Baseline / REPA.

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PCA feature maps & generation samples (baseline / REPA / REPA-Σ).

Comprehensive Cars

PCA feature maps & generation samples.

Full Report

Full report — link to be added